

Raven Biometrics

A Governed, Continuous Biometric Authentication Harness for High-Stakes
Information Environments

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Executive Summary

Biometric authentication has become the front door to the most sensitive information systems in defense, finance, healthcare, and critical infrastructure. Yet most biometric systems answer only one question: does this sample match the enrolled user, yes or no. In high-stakes information environments that binary is not enough, and the data it depends on is rarely governed to the standard those environments require.

Raven Biometrics is a governed, continuous authentication harness. It makes two contributions.

First, it decomposes the match decision into two physiological components, an identity distance and a stress distance, so it can separate three distinct states that a binary system collapses into one: a genuine user, a genuine user acting under duress or coercion, and an impostor.

Second, it treats biometric data as a governed asset rather than a disposable authentication primitive. Consent is enforced at the data layer, every decision is written to an append-only and tamper-evident audit trail, authentication events are cryptographically signed, and enrolled subjects can be fully removed on request.

Raven is delivered as a harness. The biometric sensor sits behind a single stable interface, so the difficult and durable work, governed decisioning, lives independently of any particular piece of hardware. A synthetic sensor exercises the full pipeline today; a production sensor drops in behind the same interface without changing anything downstream.

This paper describes the problem, the design principles, the layered architecture, the identity and stress decomposition, the governance model, the biometric data-management model, an evaluation, and an honest account of the system's current scope.

1. The Problem: Identity in High-Stakes Information Environments

Three assumptions sit under most deployed biometric systems, and all three break down where the stakes are highest.

The decision is treated as binary. A conventional matcher produces a similarity score and compares it to a threshold: accept or reject. That model has no vocabulary for the failure mode that matters most in a high-stakes environment, the legitimate, fully-credentialed user who is being coerced. A binary matcher waves that user straight through, because the person is genuinely who they claim to be. The credential is valid. The situation is not.

The data is treated as a primitive, not an asset. Biometric identifiers are the least revocable data class in existence. A compromised password is reset in seconds; a compromised biometric template is a permanent exposure, because a person cannot be reissued a new heartbeat, gait, or sweat chemistry. Despite this, most systems capture, store, and match biometric data with far weaker governance than they apply to ordinary records: no consent lineage, no tamper-evident record of why a decision was made, and no reliable path to delete a subject entirely. In regulated environments this is not a gap, it is a liability, and frameworks such as BIPA and GDPR treat it as one.

Liveness and provenance are afterthoughts. Presentation attacks, replayed captures, and spoofed sensors are handled, when they are handled at all, as a bolt-on rather than a first-class stage with its own audit record.

The result is a class of systems that are strong at the narrow task of matching and weak at the broader task that high-stakes information technology actually requires: proving, on demand and after the fact, who was granted access, whether they were acting freely, where the underlying data came from, and that none of it was tampered with.

2. Design Principles: Why a Harness

Raven is built as a harness rather than a device, and that choice follows from a single observation. In biometric systems, sensor hardware is the fast-moving, commoditizing layer, while governed decisioning is the durable, defensible one. Coupling the two produces a product that has to be rebuilt every time the hardware changes. Separating them produces a platform.

Five principles follow.

Modality-agnostic capture. Every sensor, synthetic or hardware, satisfies one capture contract. The pipeline downstream is agnostic to where the sample came from.

Governed decisioning is the core. The identity and stress decomposition, the consent enforcement, the audit chain, and the signing all live above the sensor interface, so they carry across hardware generations and modalities.

Fail closed. A sensor fault, a failed liveness check, a duress signal, or an identity mismatch all resolve to denied. The system never fails open.

Provenance first. Every sample, decision, and administrative action carries its source and timestamp and is written to a record that cannot be silently altered.

Honest scope. The system states plainly what is real and what is demonstrative. In a field crowded with overstatement, verifiable modesty is a competitive advantage.

3. System Architecture

Raven processes each authentication through a fixed sequence of layers. Every layer is isolated, independently testable, and writes to the governed store.

Capture. A biometric sensor returns a sample: a set of physiological and sweat-chemistry channels, a capture-quality value, and provenance. This is the only layer that touches hardware.

Liveness. Before matching, the capture is screened for quality, freshness, and replay. A low-quality, stale, or duplicated sample is denied before it can influence a decision. In production this layer also carries hardware anti-spoofing.

Template match. Enrollment builds a statistical template, a per-channel mean and standard deviation, from a subject's calm samples. Matching scores a new capture as a normalized distance from that

template, expressed in standard deviations. The score is split into two components, described in Section 4.

Decision. The two distance components resolve to one of three outcomes: genuine, duress, or impostor.

Actuation. A lock actuator grants access only on a genuine decision from a live capture on a healthy device. Every other path is denied.

Governance and audit. The sample, the signed decision, any anomaly, and the administrative events are persisted. This layer is described in Section 5.

An anomaly monitor observes the stream across authentications and raises a single event per sustained pattern, for example a run of impostor attempts or a persistent duress condition, which is the signal that matters for insider-threat and replay analysis.

4. The Differentiator: Identity and Stress Decomposition

The central idea in Raven is that a single match score throws away information that high-stakes environments need. Raven keeps it.

The biometric channels are partitioned into two groups. The identity channels are the stable, person-specific signals that vary little under acute stress: core temperature and the individual sweat-metabolite profile. The stress channels are the acute-response signals: heart rate, respiration, electrodermal activity, and cortisol. Raven scores the distance from the enrolled template separately for each group, yielding an identity distance and a stress distance.

Those two numbers resolve a decision that a single score cannot:

Identity distance	Stress distance	Decision	Interpretation
Low	Low	GENUINE	The enrolled person, acting normally. Access granted.
Low	High	DURESS	The enrolled person, but physiologically elevated. Consistent with coercion or distress. Access denied and flagged.
High	Any	IMPOSTOR	A different person. Access denied.

This is the capability a binary matcher cannot provide. A conventional system sees a coerced legitimate user and an unstressed legitimate user as identical, because both match the template. Raven sees that the identity still matches while the physiology has departed from the enrolled baseline, and it treats that departure as a first-class security event rather than a successful login.

The same mechanism delivers continuous authentication and duress detection from one primitive. There is no separate duress subsystem to maintain, no panic code to remember, and no additional user action. The signal is the user's own physiology measured against their own enrolled baseline.

Capability	Conventional biometric auth	Raven Biometrics
Match decision	Binary accept or reject	Three-state: genuine, duress, impostor
Coerced legitimate user	Granted access	Detected and denied
Continuous authentication	Rare, usually a separate system	Native, every capture
Consent lineage	Typically absent	Enforced at the data layer
Audit trail	Application logs, editable	Append-only, hash-chained, tamper-evident
Decision integrity	Unsigned	HMAC-signed per event
Subject deletion	Manual, incomplete	Right-to-be-forgotten, deletion itself audited
Hardware coupling	Tight	Behind a single sensor interface

5. Governance and Compliance by Design

Raven treats biometric data as a governed asset from the first capture. Four mechanisms make the data platform audit-ready.

Consent enforced at the data layer. A subject cannot be enrolled or matched without an active consent record that carries a purpose and a timestamp. Consent is a precondition, checked in the store, not a policy note in a document.

Append-only, hash-chained audit. Every material event, enrollment, run, anomaly, consent grant, and deletion, is written to an audit log in which each entry commits to the hash of the entry before it. Any later edit to the history breaks the chain, and a single verification pass detects it. The question "why did the system grant or deny this" becomes a query with a verifiable answer.

Signed authentication events. Each decision is signed with a device key using HMAC, so a persisted decision is tamper-evident and attributable to the device that made it.

Right to be forgotten. A subject can be fully purged: samples, template, consent, and events. The deletion is itself recorded in the audit chain, so the system can prove both that the data is gone and that its removal was authorized and logged. This is the difference between a system that merely collects biometric data and one that can be deployed where deleting it correctly is a legal requirement.

Together these turn a biometric matcher into something an auditor, a regulator, or a court can inspect. Provenance is not a feature bolted onto Raven. For a high-stakes deployment it is the precondition for deploying at all.

6. Data Management: Raven as a Biometric Data Platform

Authentication is only half of what a high-stakes biometric program requires. The other half is managing biometric data as a first-class asset across its full lifecycle: how it is modeled, ingested, indexed, retrieved,

curated for downstream use, and retired. Raven is built as a data platform, not only a matcher, and its storage layer demonstrates that model directly.

A schema built for governance and for query. Raven separates concerns across purpose-built relations: raw captures, enrolled templates, authentication events, consent records, anomalies, and the append-only audit log. Each capture carries provenance, its source and timestamp, and is retained as a structured payload, while the fields most often queried, the subject, capture quality, decision, and distance scores, are promoted to typed, indexable columns. This hybrid keeps the platform flexible enough to evolve the capture format without a migration, while staying fast to query on the dimensions operations actually use. It is the same shape that scales from an embedded database to a relational metadata store fronting object storage.

Ingestion and retrieval as governed operations. Every capture enters through a single ingestion path, so there is no route into the store that bypasses provenance or consent. Retrieval is structured around subject and time, the two axes biometric operations query most and the two that partition cleanly for scale.

Audit-ready datasets for machine learning. A recurring failure in biometric programs is training models on data of unknown origin, consent status, or permitted use. Raven's data model is designed to prevent that. Because every sample carries provenance and every subject carries a consent record, a dataset drawn for model training can be constrained to consented, permitted-use data, reproduced later with its lineage intact, and recorded in the audit chain as to what was drawn and when. This is the difference between a pile of biometric files and an audit-ready training corpus.

Lifecycle and retention as explicit states. Enrollment, active matching, retraction, and deletion are first-class operations rather than side effects. A subject is enrolled with a consent purpose and an optional expiry, matched only while that consent is active, and fully purged when it is withdrawn or lapses, with the deletion itself recorded in the audit chain. Data minimization and retention limits, the parts of a biometric program that attract the most regulatory scrutiny, are enforced by the data layer rather than left to written policy.

The scale path is a swap, not a rewrite. The reference implementation uses an embedded relational database, so the whole model runs on a laptop. The same interface targets a networked relational store for metadata and indexing with object storage for the capture payloads, partitioned by subject and time. Because the decision logic sits above the storage interface, moving from thousands of records to millions or billions is a storage change, not a redesign.

Raven therefore treats the database as the surface a biometric program is actually judged on. It shows, in a small and inspectable form, how to hold biometric data so that it stays queryable, governed, reproducible, and disposable on demand.

7. Evaluation

Raven's decision quality is measured the way biometric systems are graded, on false-accept rate and false-reject rate across the operating threshold, with duress detection reported separately.

In the reference evaluation, an enrolled identity is scored against three sample classes: genuine captures, the same subject under simulated stress, and a distinct second subject presenting as an impostor. Across

the operating band the identity threshold cleanly separates the classes, with a wide margin in which false-accept and false-reject rates are both zero and duress is detected on every stressed capture. The trade-off behaves as expected at the extremes: an overly tight threshold raises false rejects, an overly loose one raises false accepts, and the operating point sits in the stable interval between them.

This evaluation is conducted on synthetic data, which is stated plainly. Its purpose is to validate the decision logic and the governed pipeline end to end, not to certify performance against a live population. Hardware evaluation is the natural next step and is scoped in the roadmap.

8. Honest Scope and Limitations

Raven is a working reference architecture and harness, not a certified security product, and it is described as such deliberately.

The biometric input is currently synthetic. A real-sensor interface is provided and the entire pipeline runs against it, but no hardware is integrated. The liveness and event-signing layers are functional and demonstrative rather than hardened: production deployment requires hardware anti-spoofing, cryptographic nonces, and a managed key infrastructure. The evaluation is synthetic.

None of this is hidden, because the value Raven demonstrates does not depend on it. The identity and stress decomposition, the three-state decision, and the governed, auditable data platform are real, tested, and reproducible today. The remaining work is integration, not invention.

9. Deployment and Scale

Raven is designed to run as two cooperating tiers. An edge tier performs capture, liveness, matching, and actuation close to the point of use, suitable for a wearable or an embedded reader. A governance tier holds the enrolled templates, consent records, and the append-only audit chain, and is the system of record for compliance.

The storage layer is deliberately interface-clean. The reference implementation uses an embedded database; the same interface targets a networked relational store with object storage for sample payloads, which is the path to high-volume, multi-subject deployment without changes to the decision logic above it.

Sensor integration proceeds through the existing capture contract, so bringing Raven to a specific hardware platform is an implementation of one interface rather than a redesign of the system.

10. About

Raven Biometrics is a product of Project Black Box LLC, an independent research and development company focused on measurement, governance, and adversarial resilience in information systems. It descends from the RavenLock concept first documented in 2025 and reframed here on real biometric primitives with tighter scope and a governed data platform.

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